



Translated

From: e-maxx.ru

## Number of paths of fixed length / Shortest paths of fixed length

The following article describes solutions to these two problems built on the same idea: reduce the problem to the construction of matrix and compute the solution with the usual matrix multiplication or with a modified multiplication.

### Number of paths of a fixed length

We are given a directed, unweighted graph  $G$  with  $n$  vertices and we are given an integer  $k$ . The task is the following: for each pair of vertices  $(i, j)$  we have to find the number of paths of length  $k$  between these vertices. Paths don't have to be simple, i.e. vertices and edges can be visited any number of times in a single path.

We assume that the graph is specified with an adjacency matrix, i.e. the matrix  $G$  of size  $n \times n$ , where each element  $G[i][j]$  equal to  $1$  if the vertex  $i$  is connected with  $j$  by an edge, and  $0$  if they are not connected by an edge. The following algorithm works also in the case of multiple edges: if some pair of vertices  $(i, j)$  is connected with  $m$  edges, then we can record this in the adjacency matrix by setting  $G[i][j] = m$ . Also the algorithm works if the graph contains loops (a loop is an edge that connect a vertex with itself).

It is obvious that the constructed adjacency matrix is the answer to the problem for the case  $k = 1$ . It contains the number of paths of length  $1$  between each pair of vertices.

We will build the solution iteratively: Let's assume we know the answer for some  $k$ . Here we describe a method how we can construct the answer for  $k + 1$ . Denote by  $C_k$  the matrix for the case  $k$ , and by  $C_{k+1}$  the matrix we want to construct. With the following formula we can compute every entry of  $C_{k+1}$ :

$$C_{k+1}[i][j] = \sum_{p=1}^n C_k[i][p] \cdot G[p][j]$$

It is easy to see that the formula computes nothing other than the product of the matrices  $C_k$  and  $G$ :

$$C_{k+1} = C_k \cdot G$$

Thus the solution of the problem can be represented as follows: